Adding & Subtracting Polynomials — Practice Worksheet

Name: ______________________ Date: ______________ Period: ____________

Type 1 — Adding Polynomials (Find the Sum)

EXAMPLE — Look at this before you begin:

Find the sum: (5ab + 5ac - 9bc) + (-ab + 10ac + 4bc)
Group like terms: (5ab - ab) + (5ac + 10ac) + (-9bc + 4bc)
Simplify: 4ab + 15ac + -5bc
Answer: 4ab + 15ac - 5bc. Show all work!

Find each sum. Show all work. Circle your final answer.

Problem 1-A. Find the sum.

$$(3xy + 2xz - 4yz) + (-xy + 5xz + 2yz)$$

Problem 1-B. Find the sum.

$$(7ab - 3ac + bc) + (2ab + 6ac - 5bc)$$

Problem 1-C. Find the sum.

$$(4m^2n - 3mn^2 + 2mn) + (-m^2n + 5mn^2 - 7mn)$$

Problem 1-D. Find the sum.

$$(8x^2y + 3xy^2 - xy) + (-2x^2y - xy^2 + 4xy)$$

Problem 1-E. Find the sum.

$$(-5a^2b + 4ab^2 - 3ab) + (3a^2b - 2ab^2 + 8ab)$$

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Type 2 — Subtracting Polynomials (Find the Difference)

EXAMPLE — Look at this before you begin:

Find the difference: $(-12r^2s + 2rs^2 + 6rs) - (3r^2s + 8rs^2 - rs)$

Distribute the negative: $-12r^2s + 2rs^2 + 6rs - 3r^2s - 8rs^2 + rs$

Combine like terms: $-15r^2s - 6rs^2 + 7rs$

Answer: $-15r^2s - 6rs^2 + 7rs$. Show all work!

Find each difference. Show all work. Circle your final answer.

Problem 2-A. Find the difference.

$$(6x^2y - 3xy^2 + 5xy) - (2x^2y + xy^2 - 3xy)$$

Problem 2-B. Find the difference.

$$(8ab^2 - 5a^2b + 3ab) - (-2ab^2 + 4a^2b + ab)$$

Problem 2-C. Find the difference.

$$(5m^2n - 3mn + 2mn^2) - (2m^2n + mn - 4mn^2)$$

Problem 2-D. Find the difference.

$$(-4r^2s + 7rs^2 - 2rs) - (3r^2s - 2rs^2 + 5rs)$$

Problem 2-E. Find the difference.

$$(9x^2z - 4xz^2 + 6xz) - (-x^2z + 2xz^2 - 3xz)$$

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Type 3 — Polynomial Perimeter Word Problems

EXAMPLE — Look at this before you begin:

Find the perimeter of a triangle with sides $(2m+5)$, $(10m^2-8)$, and (m^2+8m).

$P = (2m+5) + (10m^2-8) + (m^2+8m)$

Combine like terms: $(10m^2+m^2) + (2m+8m) + (5-8)$

Answer: $11m^2 + 10m - 3$. Use P = Side + Side + Side.

Find each perimeter. Show all work. Circle your final answer.

Problem 3-A.

Find the perimeter of a triangle with sides $(3x + 2)$ in., $(5x - 1)$ in., and $(2x + 4)$ in.

$P = (3x + 2) + (5x - 1) + (2x + 4)$

Problem 3-B.

Find the perimeter of a triangle with sides $(m^2 + 2m)$ ft, $(3m^2 - 5)$ ft, and $(2m + 7)$ ft.

$P = (m^2 + 2m) + (3m^2 - 5) + (2m + 7)$

Problem 3-C.

Find the perimeter of a triangle with sides $(2n^2 - 3)$ cm, $(n^2 + 4n)$ cm, and $(3n + 6)$ cm.

$P = (2n^2 - 3) + (n^2 + 4n) + (3n + 6)$

Problem 3-D.

Find the perimeter of a triangle with sides $(4a + 3)$ m, $(2a^2 - a)$ m, and $(a^2 + 5a + 1)$ m.

$P = (4a + 3) + (2a^2 - a) + (a^2 + 5a + 1)$

Problem 3-E.

Find the perimeter of a triangle with sides $(5k^2 - 2k)$ yd, $(3k + 7)$ yd, and $(2k^2 + k - 3)$ yd.

$P = (5k^2 - 2k) + (3k + 7) + (2k^2 + k - 3)$

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Type 4 — Real-World Polynomial Applications

EXAMPLE — Look at this before you begin:

Revenue: $(15x^3 + 6x^2 + 700)$ dollars. Cost: $(60x^2 + 2000)$ dollars.
Profit = Revenue - Cost
$= (15x^3 + 6x^2 + 700) - (60x^2 + 2000)$
$= 15x^3 - 54x^2 - 1300$. Then substitute the given x to find profit.

Write a simplified profit expression, then evaluate. Show all work.

Problem 4-A.

The revenue for a company selling x products is modeled by $(8x^3 + 4x^2 + 300)$ dollars. The operating cost is modeled by $(3x^2 + 800)$ dollars. Write a simplified expression for the profit. Then find the profit if $x = 5$.

Profit expression: ______________________ **If x = 5, profit =** ______________________

Problem 4-B.

The revenue for a company selling x products is modeled by $(12x^3 + 8x^2 + 400)$ dollars. The operating cost is modeled by $(50x^2 + 1200)$ dollars. Write a simplified expression for the profit. Then find the profit if $x = 8$.

Profit expression: ______________________ **If x = 8, profit =** ______________________

Problem 4-C.

The revenue for a company selling x products is modeled by $(20x^2 + 5x + 900)$ dollars. The operating cost is modeled by $(8x^2 + 600)$ dollars. Write a simplified expression for the profit. Then find the profit if $x = 4$.

Profit expression: ______________________ **If x = 4, profit =** ______________________

Problem 4-D.

The revenue for a company selling x products is modeled by $(6x^3 + 10x^2 + 200)$ dollars. The operating cost is modeled by $(2x^2 + 750)$ dollars. Write a simplified expression for the profit. Then find the profit if $x = 6$.

Profit expression: ______________________ **If x = 6, profit =** ______________________

Problem 4-E.

The revenue for a company selling x products is modeled by $(15x^3 + 6x^2 + 700)$ dollars. The operating cost is modeled by $(60x^2 + 2000)$ dollars. Write a simplified expression for the profit. Then find the profit if $x = 10$.

Profit expression: ______________________ **If x = 10, profit =** ______________________

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Answer Key

Type 1 — Adding Polynomials

1-A $2xy + 7xz - 2yz$
1-B $9ab + 3ac - 4bc$
1-C $3m^2n + 2mn^2 - 5mn$
1-D $6x^2y + 2xy^2 + 3xy$
1-E $-2a^2b + 2ab^2 + 5ab$

Type 2 — Subtracting Polynomials

2-A $4x^2y - 4xy^2 + 8xy$
2-B $10ab^2 - 9a^2b + 2ab$
2-C $3m^2n - 4mn + 6mn^2$
2-D $-7r^2s + 9rs^2 - 7rs$
2-E $10x^2z - 6xz^2 + 9xz$

Type 3 — Perimeter Word Problems

3-A $P = 10x + 5$
3-B $P = 4m^2 + 4m + 2$
3-C $P = 3n^2 + 7n + 3$
3-D $P = 3a^2 + 8a + 4$
3-E $P = 7k^2 + 2k + 4$

Type 4 — Real-World Applications

4-A Profit = $8x^3 + x^2 - 500$; at x=5: \$525
4-B Profit = $12x^3 - 42x^2 - 800$; at x=8: \$2,656
4-C Profit = $12x^2 + 5x + 300$; at x=4: \$512
4-D Profit = $6x^3 + 8x^2 - 550$; at x=6: \$1,034
4-E Profit = $15x^3 - 54x^2 - 1300$; at x=10: \$8,300